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How do text characteristics impact user engagement in social media posts: Modeling content readability, length, and hashtags number in Facebook

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A R T I C L E I N F O

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A B S T R A C T

This study examines whether text characteristics in branded Facebook image posts associate with consumer engagement and brand awareness. The examined text characteristics include i) readability indices, ii) text length, and iii) number of hashtags. A dataset of 135 image posts with description texts was exported from a Fashion retail Facebook business page providing post performance metrics in terms of engagement (expressed in likes) and awareness (expressed in reaches and impressions). Positive associations were indicated between all performance metrics and the text's length, as well as the number hashtags. The readability index of Gunning Fog revealed strong associations with both engagement and awareness, while the Flesch Kincaid reading ease index was associated only with awareness metrics of reaches and impressions. Overall, the results revealed that, the posts' text which is easy to read, long (more than 31 words, or more than 321 characters), and contains many hashtags tends to achieve higher performance of engagement and awareness. This research contributes to prior literature by shedding light on the role of text characteristics of branded messaging in social media and offering insights for brand communication and social media message strategies.

1. Introduction

The rise of social media provides firms with a new avenue to connect with their targeting customers and achieve higher levels of consumer-brand engagement. Consumer-brand engagement in social media is tightly linked with increased conversion and purchase intention (Consuegra, Faraoni, Díaz & Ranfagni, 2018; Eslami, Ghasemaghaei & Hassanein, 2021). Consumer engagement in social media brand communities also helps to build brand trust, and brand loyalty (Bianchi & Andrews, 2018; Zhang, Lu, Torres & Chen, 2018). Researchers agree that understanding consumer-brand engagement behavior within social media is fundamental for the performance of retail firms, especially of small and medium enterprises (SMEs), including the Fashion industry (Hsiao, Wang, Wang & Kao, 2020; Kachba, Plath, Ferreira & Forcellini, 2012).

The Fashion industry has been dramatically affected by social media since it provides a useful channel to disseminate the design of independent designers and fashion firms (Kachba et al., 2012). Despite the importance of fashion brands in industrial marketing, there is a gap in the literature as regards social media marketing strategies and consumer engagement in the fashion industry (Hsiao et al., 2020). Overall, little is still known about how consumers engage with retail brands through social media (Bianchi & Andrews, 2018; Santos, Cheung, Coelho & Rita, 2022), and how branded messaging characteristics affect consumer engagement (Davis, Horváth, Gretry & Belei, 2019).

Although previous researchers have examined the potential effects of a wide of set of social media post characteristics on user engagement, these can vary according to the industry sector and yet there are more attributes to be considered. For example, previous studies examined the way that different types of posts (images, videos, links) can affect user engagement in social media (Dhaoui & Webster, 2021), the role of the posting timing (weekday or hour) or the time interval between posts (Fournier & Avery, 2011; Schulze, Schöler & Skiera, 2014), the emotional as opposed to informational appeal (Molina, Gómez, Lyon, Aranda & Loibl, 2020), the brand's responsiveness (e.g. Proserpio & Zervas, 2017), the level of interactivity and the vividness of the message (Sreejesh, Paul, Strong & Pius, 2020), and the cocreation of content expressed in number of likes, shares, and comments (Eslami et al., 2021). As noticed, little research is conducted on the role of readability and other text-related characteristics of the social media posts, despite

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evidence on their significant effects on social media user engagement (Ismail, Kuppusamy, Kumar & Ojha, 2019) and perceived familiarity with the promoted brand (Davis et al., 2019).

Motivated by the above-mentioned research gaps and the role of Readability on user engagement and brand familiarity, this study suggests that the Facebooks posts' text-related characteristics might be associated with customer engagement and brand awareness in fashion industry. The main objective of this study is to explore the role of Facebook post's text readability features (including readability indices, text length, and number of hashtags), on a set of user engagement and brand awareness metrics (including metrics of likes, reaches, and impressions). The study focuses on image posts (excluding videos and links) since they are representative in the business sector of Fashion industry. Towards this goal, the research questions (RQs) are formed as follows:

- RQ1: Are there any significant associations between text readability features of branded image posts and customer engagement?
- RQ2: Are there any significant associations between text readability features of branded image posts and brand awareness?

The study is conducted on a dataset extracted from a Facebook women Fashion business page, in Greece. The dataset covers a six-month date range and includes organic (i.e., not paid) image posts containing text description. The research results contribute to the deeper understanding of user behavior in social media branded messages, and the factors that affect consumer engagement and brand awareness in the segment of Fashion industry. This study also contributes to the readability literature, by challenging the paradigm that easier-to-read messages result in more positive consumer behavior.

2. Literature review

2.1. Readability features and indices

Readability has long been used to predict the difficulty people will have in understanding written content. Readability is defined as "the ease of understanding or comprehension due to the style of writing" (Klare, 1963, p. 15). An earlier definition of readability was "the extent to which readers understand a text, read it at an optimal speed, and find it interesting" (Dale & Chall, 1949, p. 19).

Nowadays it is well established that the readability level of texts is a crucial factor regarding their appropriateness for ages groups and/or audiences. Readability features include several linguistic levels of text like level of vocabulary, word length, word ambiguity, figurative language, sentence length, syntactic complexity, cohesion, and number of paragraphs (Temnikova, Vieweg & Castillo, 2015).

In websites' Search Engine Optimization, readability is usually reflected in the length of paragraphs, the length of sentences, the use of passive or active voice, the use of transition words, the subheading distribution, the use of consecutive phrases, and the scores of readability indexes. Readability indexes are used to determine the comprehension difficulty of the material (Flesch, 1948) and avoid needless complexities in the mechanics of writing (Gunning, 1969).

The most common indexes to measure readability of a text are the Flesch–Kincaid (FK), and the Gunning fog index (GFI) (Ismail et al., 2019). The GFI estimates the number of years of schooling a person needs to understand a given text on the first reading (Cortés, Rivera & Carbonelld, 2022). The FK index estimates the grade level or number of years of education required for someone to understand the information, considering the average sentence and word length to assess (Wang, Miller, Schmitt & Wen, 2013).

2.2. Readability and user engagement in social media posts

Researchers explain the positive effects of readability on user engagement, because of the perceived processing fluency (Rennekamp, 2012). As explained, consumers respond more positively to messages that feel easier to process (Lee & Aaker, 2004; Lee & Labroo, 2004) and hence, easy-to-read messages result in greater consumer engagement with a brand.

So far, most research on readability is mainly focused on long texts (online content and educational material), while the readability of short messages receives little attention (Davis et al., 2019), and empirical research into the readability of brand communication is lacking.

Previous research on social media readability has mainly examined the tweets' specific features (like length limitations, use of hashtags, and emojis) that can affect readability and make messages difficult to read (Davenport & DeLine, 2014; Temnikova et al., 2015). Previous literature has also showed that text characteristics, including readability, significantly affect user engagement in websites (Ismail et al., 2019); however, the associations between readability and user engagement or brand awareness in social media is significantly under researched. A recent study of Davis et al. (2019) showed that readability and text features of hashtags and at-mentions are positively associated with user engagement in Twitter messages (tweets). Readability is also linked with the users' perceived familiarity towards a brand (Davis et al., 2019), hence we argue that readability is associated not only with consumer engagement but also with brand awareness.

2.3. Social media post characteristics and consumer brand engagement

Recently, social media data mining techniques have traced new paths towards understanding and predicting user behavior in social media platforms (e.g., Colladon, Gloor & Iezzi, 2020; Dong, Liao & Zhang, 2018; Kushwaha & Kar, 2020a; Kushwaha et al., 2020b, Kushwaha et al., 2020c; Wu, Li, Shen & He, 2020). As fact, sentiment analysis and social-media analytics have revealed useful insights integrated in the marketing and communication strategies of the SMEs worldwide.

Despite its essential role on social media marketing, consumer brand engagement lacks research and conceptualization. According to Santos et al. (2022) consumer brand engagement reflects the interaction with brand communities on social media. This kind of engagement goes beyond simple engagement. The brand community involves the brand, its products, the firm, and other consumers, so the engagement with the brand community reflects a broader process of interaction.

Previous studies have explored a large set of social-media data analytics to interpret the factors that tend to engage consumers and increase conversions. Recently, Eslami et al. (2021) showed that the cocreation of content in social media, expressed in numbers of interactions (number of likes, and shares) can lead to increased user engagement their findings revealed no association between consumer engagement and the number of comments. Similarly researchers found that the use of hashtags attracts user attention and generates higher interactivity with the content (Cheng, Lam & Chiu, 2020; Salomon, 2013). Therefore, based on the numbers of likes and shares per post, the use of eye-catching hashtags, marketers and decision makers aim to increase branding by defining which and why product or service post is more engaging.

Researchers also observed that most engaging interactions came from image posts rather than any other post type like status, link, or video (Cvijikj, Spiegler & Michahelles, 2011; Dhaoui & Webster, 2021; Khan, Hafeez, Ijaz & Shaheen, 2019). Complementary, links containing information about cultural, and children's activities are proved to increase interactivity in Facebook pages (Molina et al., 2020). Saturdays and Fridays are more engaging than weekdays, and nighttime posting is more engaging than daytime posting (Fournier & Avery, 2011; Khan et al., 2019; Schulze et al., 2014). However, contradictory findings (Cvijikj et al., 2011) support that posting are not associated with consumer engagement.

User engagement is social media can vary according to the message format and message content as well (Leung, Bai & Erdem, 2017). For instance, Molina et al. (2020) examined the format of Facebook posts and found that emotional appeal in the content achieves higher interaction, compared to informational content.

Table 1
Facebook business page insights: raw dataset.

Exported feature	Number of records
Total page likes	1756
Total posts likes	3690
Average page likes	1688
Total page followers	1173
Average organic post reaches	473
Average paid post reaches	5985
Average post reactions	23
Average post comments	0
Average post shares	1
Other (hide, report as spam, and unlikes)	3

Note. Time period: 30/4/2020 - 25/10/2020.

In the context of Fashion industry, research revealed there are four factors that shape consumer brand engagement. These factors include i) interaction, information sharing, and opinion exchanging among social media platforms, ii), electronic word of mouth (e-WoM) affected by testimonials, reviews, comments, and ratings, iii) entertainment using social media platforms increase user stimulation, and iv) the actual user experience determines the outcome (Antoniadis, Xanthakou & Assimakopoulos, 2019).

Today, the online fashion industry is significantly affected by the social media marketing (Hsiao et al., 2020; Posner, 2015). Fashion brand credibility through social image positively affects brand image, purchase intention and purchase intention (Sweeney & Swait, 2008). Social media ads significantly affect consumers choice of clothes brands, while word of mouth and reviews have also a significant effect on choosing a brand (Davidaviciene, Davidavičius & Tamosiuniene, 2019). Today, fashion product customers prefer to buy online since they witness facilities like easy browsing, product returns, on-time delivery, coupons, discounts, offers, etc. (Pattnaik & Trivedi, 2020). Moreover, social media marketing proves to be effective and quite successful tool in the hands of fashion industry when it comes to affect and increase sales and awareness among targeted consumers (Pattnaik & Trivedi, 2020).

Undoubtedly, social media marketing has a significant impact on customers perceived brand value (Consuegra et al., 2018) and customer-brand engagement in the Fashion industry. Exploiting previous findings on social media post characteristics and further exploring the role of readability on consumer brand engagement can reveal significant insights on consumer research.

3. Methodology

3.1. Facebook business page and data extraction

The Facebook data were extracted from a Fashion Facebook Business Page, of a Greek woman fashion retail online shop. The Fashion store sells women fashion products through both physical and online store. The current business manages to promote its products using social media, especially Facebook and Instagram. The store promotes more than 50 women fashion clothing brands. The store's social media business pages are currently followed by 1800 followers on Facebook and 3290 followers on Instagram.

The dataset was exported through the Facebook Business Manager, were the authors having administration access could directly export the page statistics in csv format. The Facebook data was gathered in a six-month period of 180 days, from 30th of April 2020 to 25th of October 2020. The total number of the exported published posts was 135. The page active audience was mainly women. Overall, female users expressed higher engagement behavior than males (for instance the average number of likes for females was 212 while for males it was only 56), and most of the engaged users were between 25 and 44 years old.

Table 1 depicts the exported data for the examined period.

Table 2
List of examined post's text characteristics.

Feature	Description
Flesch Kincaid reading ease	Readability ease score
Gunning Fog Index	Readability ease score
Word count	Number of words per post
Character count	Number of characters per post
Hashtag count	Number of hashtags per post

3.2. Data processing and analysis

Data was pre-processed and cleaned using Microsoft Excel. First outliers and very small numbers of collected data including the number of shares, number of comments, average post reactions and other (hide, report as spam, and unlikes), were removed. Then the readability metrics were calculated (See 3.2.1) and the performance metrics were grouped into engagement and awareness metrics (See 3.2.2.) The statistical analyses were conducted in SPSS software. Because of the non-normal distribution (p<0,05) observed in the dataset (Shapiro & Wilk, 1965), non-parametric methods were applied to test the significant differences and associations between the pairs of the examined variables. The Kruskal-Wallis Test was applied to examine the associations between performance and the readability metrics of Text readability indices and Text length. The method was chosen because Text readability indices and Text length was categorized into distinct categories (See 3.2.1), hence significant differences in performance should be examined between the defined groups. The Spearman rank correlation test was applied to examine ethe associations between performance metrics and the number of hashtags, since the number of hashtags was expressed in continuous variable and no categorization was applied.

Following we describe the calculation methods of text readability and Facebook performance. Readability metrics include a) Text readability indices, b) Text length, and c) Number of hashtags, and Facebook performance metrics are categorized into engagement and awareness metrics.

3.2.1. Text readability metrics

The text related features were all calculated and included the following metrics: a) Text readability indices, b) Text length, including word count and character count, and c) Number of hashtags. The study selected to apply the readability indexes of Flesch–Kincaid reading ease, and Gunning fog index, because they are the most common indexes to measure readability of online text (Ismail et al., 2019).

Table 2 summarizes all the examined metrics of the post' text characteristics

The Flesch Kincaid reading ease and the Gunning Fog Index algorithms were used to measure readability levels. The readability indices scores were measured via the WebFX Readability Test online tool (WebFX, 2020).

The Flesch Kincaid reading ease classifies text scores according to the following segmentation. "Very difficult" to read text scores from 0 to 30, "Difficult" to read text from 30 to 50, "Fairly difficult" to read from 50 to 60, "Easily understood" text from 60 to 70, "Fairly easy to Read" text from 70 to 80, "Easy to Read" text from 80 to 90, and "Very Easy to Read" text from 90 to 100 (Eleyan, Othman & Eleyan, 2020; Jackson, 2020). The Flesch Kincaid higher scores indicated text that is easy to understand, while lower numbers mark increased difficulty. The algorithm considers the core measures of word length and sentence length based on the following formula (Flesch, 1998):

$$206.835 - 1.015\ [(\text{total words}/\text{total sentences}) + 100(\text{total syllabes}/\text{total words})] \quad (1)$$

The Gunning Fog Index measures the text readability indicating the level of education needed to understand each level. The Gunning fog

Table 3
Flesch Kincaid reading ease categories.

Flesch Kincaid reading ease							
Score	Very difficult	Difficult	Fairly difficult	Easily understood	Fairly easy	Easy	Very easy
Values	−3	−2	−1	0	1	2	3

Table 4
Gunning fog index categories.

Gunning fog index										
Score	6	7	8	9	10	11–12	13–15	16	17–20	20+
Values	1	2	3	4	5	6	7	8	9	10

index scale scores 6 for 6th grade level, 7 for 7th grade level, 8 for 8th grade level, 9 for high school freshman level, 10 for high school sophomore level, 11–12 for high school senior, from 13 to 15 for college junior, sophomore, freshman level, 16 for college senior level, from 17 to 20 for post-graduate level, and for greater than 20 for post-graduate plus level (Eleyan et al., 2020). Generally, texts requiring near-universal understanding generally need an index less than 8. The computed result of the Gunning Fog Index algorithm is based on a formula considering the average sentence length and the number of the complex words (consisting of three or more syllables), as follows (Gunning, 1969):

$$0.4\ \left[(\text{words/sentences})\ +\ 100(\text{complex words/words})\right] \tag{2}$$

As regards the text length we categorized text according to the number of characters and the number of words. The categorization of character- length was based on the criteria suggested by Jackson (2020), who showed that posts containing more than 80 characters tend to achieve higher performance of user engagement performance. Hence, the authors taking into consideration that 80 characters distinguishes uses engagement they decided to classify the texts based on 80 characters sets 0–80, 81–160, 161–240, 241–320, 321–400, 400–480, 481–560. Based on (Jackson, 2020; Lee, 2020) the ideal number of words for headlines is from 5 to 6 words. The authors decided to set the range of 5-word post text scale. Finally, the number of hashtags per post is the actual classification criterion. Categorization includes features that characterize the post based on the type of product or service post (e.g., image).

Tables 3 and 4 show how the authors coded the reading ease using numbers, while Tables 5 and 6 show the calculation of the text length metrics based on the number of characters and words.

3.2.2. Performance metrics

Performance metrics measure how effective the post is in terms of user engagement and page/brand awareness and include metrics such as likes, shares, reaches, impressions. Engagement includes performance metrics of liking, while awareness includes performance metrics of reaches and impressions. While engagement metrics are triggered directly by the users' interaction with the post, awareness performance metrics express a more passive consumer behavior (expressed in views) and are handles by the Facebook algorithm.

The term "reaches" is the total number of people who see the post. The term "impressions" refers to the number of times the post is displayed, no matter if it was clicked or not and the term "engagements," define user actions on the post when at the same time impressions are not able to provide enough evidence for distinguishing whether the user has paid attention to the post or not (Keyhole, 2020).

The Facebook post performance metrics include "Post shares", "Post likes", "Lifetime post total reaches", "Lifetime post total impressions", "Lifetime post engaged users", "Lifetime post impressions by people who have liked the page", "Lifetime post reaches by people who like the page", and "lifetime people who have liked the page". However, as mentioned in data analysis, this study excludes the metric of post share since the extracted dataset did not provide with an efficient amount of post shares ($n = 1$). The list of the examined performance metrics is presented in Table 7.

4. Results

This section presents the descriptive statistics of the examined features, and all the results of the statistical analyses conducted between the readability metrics and the examined performance metrics (engagement and awareness). In the end, we present the main characteristics of the most engaged metrics, to confirm and discuss the quantitative findings.

4.1. Descriptive statistics

Table 8 presents the descriptive statistics of the calculated readability features and the exported Facebook performance metrics for the examined period.

4.1.1. Significant differences in posts' performance metrics based on text readability indices and length

Tables 9 and 10 depict the significant differences in engagement performance scores among the examined categories of. readability indices: Flesch Kincaid and Gunning fog index.

As depicted the readability score of Flesch Kincaid is significantly associated with the awareness metric of lifetime post impressions., while The Gunning fog index reveals significant differences in both engagement and awareness metrics as depicted in Table 10. As derived from the descriptive statistics results, the highest value of Lifetime Post Impressions by People who Have Liked your Page, were noticed in the level "Easy to Read" of the Flesch Kincaid ranges. Whereas the lowest engagement values were observed in the range 'Difficult'.

Tables 11 and 12 depict the differences in engagement and awareness performance scores across different categories of text length based on the number of words and characters. As depicted, the length of the text is associated with all engagement and awareness performance metrics. In particular, the posts containing 31–55 words, or 321–580 characters achieved the highest scores throughout all the Facebook post performance metrics, while the lowest performance was achieved by the posts ranging from 6 to 15 words or 0–160 characters length.

4.1.2. Correlation analysis between the number of hashtags and post's performance metrics

The Spearman rank correlation was used to measure the degree of association between the variable of hashtag and the Facebook performance metrics. As depicted in Table 13, the number of hashtags in an image post is positively associated with all engagement and awareness metrics of likes, reaches and impression.

4.2. Characteristics of the top 3 most engaging facebook posts

To evaluate the above presented quantitative results we seek to explore their application in the examined dataset of posts. In this study we examine the engagement performance separately for user engagement for "Lifetime post engaged users" and for "Lifetime people who have liked the page and engaged with the post". This happened because people who like a page receive more posts from that page (The Design Space, 2020). The user engagement (E) is measured by the fraction of

Table 5
Character number categories.

Character number							
Score	0–80	81–160	161–240	241–320	321–400	400–480	481–560
Values	1	2	3	4	5	6	7

Table 6
Word count categories.

Word count											
Score	0–5	6–10	11–15	16–20	21–25	26–30	31–35	36–40	41–45	46–50	51–55
Values	1	2	3	4	5	6	7	8	9	10	11

Table 7
List of Facebook performance metrics.

Feature	Performance type	Description
Post likes	Engagement	The number of times a post was liked by Facebook users
Total likes	Engagement	The total number of times posts were liked by Facebook users
Lifetime people who have liked the page and engaged with the post	Engagement	The number of people who have liked your page and clicked anywhere in your posts. (Unique users)
Lifetime post engaged users	Engagement	The number of Unique people who engaged in certain ways with your page post, for example by commenting on, liking, sharing, or clicking upon elements of the post. (Unique users)
Lifetime post total reaches	Awareness	The number of people who had your page's post enter their screen. Posts include statuses, images, links, videos and more. (Unique users)
Lifetime post total impressions	Awareness	The number of times your Page's post entered a person's screen. Posts include statuses, images, links, videos and more. (Total count)
Lifetime post impressions by people who have liked the page	Awareness	The number of impressions of your page post to people who have liked your page. (Total count)
Lifetime post reaches by people who like the page	Awareness	The number of people who saw your page post because they have liked your page (Unique users)

Table 8
Descriptive statistics.

	N	Min	Max	Mean	Std. Dev.
Flesch Kincaid reading ease (FK)	135	−3	2	−2,56	,886
Gunning fog index (GF)	135	1	9	2,62	2157
Word count	135	2	11	3,89	1777
Characters count	135	1	7	2,37	1202
Hashtags count	135	0	23	4,85	5005
Post likes	135	6	112	26,95	15,778
Lifetime post total reaches	135	281	1648	747,10	245,614
Lifetime post total impressions	135	313	2005	866,98	300,638
Lifetime post engaged users	135	3	252	44,51	37,778
Lifetime post impressions by people who liked your page	135	265	1163	561,55	191,457
Lifetime post reaches by people who like your page	135	233	904	468,27	141,457
Lifetime people who have liked your page & engaged with your post	135	3	185	36,40	30,539

Table 9
Kruskal-Wallis test statistics for Flesch Kincaid reading ease.

	Post shares	Lifetime post total reaches	Lifetime post total impressions	Lifetime post engaged users	Lifetime post impressions by people who have liked your page	Lifetime post reaches by people who like your page	Lifetime people who have liked your page and engaged with your post
Kruskal-Wallis	8056	7479	7554	6361	11,180	9752	7236
df	5	5	5	5	5	5	5
Asymp. sig.	,153	,187	,183	,273	**,048***	,083	,204

* Statistical significance at $p = 0.05$ | Grouping Variable: Flesch Kincaid reading ease.

Table 10
Kruskal-Wallis test statistics for Gunning fog index.

	Post likes	Lifetime post total reaches	Lifetime post total impressions	Lifetime post engaged users	Lifetime post impressions by people who have liked your page	Lifetime post reaches by people who like your page	Lifetime people who have liked your page and engaged with your post
Kruskal-Wallis H	13,320	10,166	12,461	16,889	21,470	19,852	17,918
df	8	8	8	8	8	8	8
Asymp. sig.	,101	,254	,132	**,031***	**,006***	**,011***	**,022***

* Statistical significance at $p = 0.05$ | Grouping Variable: Gunning fog index.

Table 11
Kruskal-Wallis test statistics for “Word count”.

	Post shares	Post likes	Lifetime post total reaches	Lifetime post total impressions	Lifetime post engaged users	Lifetime post impressions by people who have liked your page	Lifetime post reaches by people who like your page	Lifetime people who have liked your page and engaged with your post
Kruskal-Wallis	2324	30,750	26,249	32,455	34,428	48,101	47,158	37,556
df	7	7	7	7	7	7	7	7
Asymp. sig.	,940	**,000***	**,000***	**,000***	**,000***	**,000***	**,000***	**,000***

*Statistical significance at $p = 0.05$ | Grouping variable: Word count.

Table 12
Kruskal-Wallis test statistics for “Characters count”.

	Post likes	Lifetime post total reaches	Lifetime post total impressions	Lifetime post engaged users	Lifetime post impressions by people who have liked your page	Lifetime post reaches by people who like your page	Lifetime people who have liked your page and engaged with your post
Kruskal-Wallis H	38,230	29,123	36,004	39,299	54,293	51,706	41,579
df	6	6	6	6	6	6	6
Asymp. sig.	**,000***	**,000***	**,000***	**,000***	**,000***	**,000***	**,000***

*Statistical significance at $p = 0.05$ | Grouping Variable: Characters count.

Table 13
Spearman correlation between the number of Hashtags and performance metrics.

		Hashtags number	Post likes	Lifetime post total reaches	Lifetime post total impressions	Lifetime post engaged users	Lifetime post impressions by people who have liked your page	Lifetime post reaches by people who like your page	Lifetime people who have liked your page and engaged with your post
Hashtags number	Correlation coefficient	1000	**,446****	**,362****	**,430****	**,506****	**,627****	**,600****	**,552****
	Sig.(2-tailed)	.	,000	,000	,000	,000	,000	,000	,000
	N	135	135	135	135	135	135	135	135

*. Correlation is significant at the 0.05 level (2-tailed).
**. Correlation is significant at the 0.01 level (2-tailed).

the total number of engaged users divided by the number reached users as depicted in formulas. It this case user engagement is measured separately for user who did or did not like the page.

Total Lifetime post engaged users’ engagement:

$$E = \frac{Lifetime\ Engaged\ Users}{Lifetime\ Post\ Total\ Reach} * 100 = 6\% \tag{3}$$

Total Lifetime people who have liked the page and engaged with the post engagement:

$$E = \frac{\text{Lifetime people who have liked the page and engaged with the post}}{\text{Lifetime post reaches by people who like the page}} * 100 = 8\% \tag{4}$$

In order to have a closer look towards the association between the users’ engagement and readability characteristics (redability indices, text lenght, number of hashtags) we provide an example of metrics of the most engaging posts of the extracted dataset. As depicted in Table 14 the top three most engaging posts are of a particular length of words (31–55) and characters (321–560). The number of hashtags mentioned in text is quite high e.g. from 17 to 23 hastags. As depicted, the redability scores of Flesch Kincaid in the first two posts are quite low, revealing the weak association between readiness ease and engagement performance in Facebook image posts. However, the readability score of Gunning fog is quite high (>8) in all three posts, conforming its stonger association with performance metrics, as opposed to the Flesch Kincaid.

Distinguishing these occasions, we try to expose the actual behavior a user might have over a post content and context hoping to provide valid information to decision makers.

Table 14
Characteristics of the top 3 most engaging Facebook posts.

Post ID	Flesch Kincaid reading ease	Gunning fog index	Word count	Characters number	Hashtags Number	Lifetime post total reaches	Lifetime post total impressions	Lifetime post engaged users
1	−38,6	8,7	54	523	23	1648	2005	252
2	−51,8	7,5	31	338	17	1516	1821	223
3	81,5	14,8	34	374	20	1440	1684	199

5. Discussion

The results of this study revealed a several significant associations between the examined performance metrics and readability features, on the exported dataset of Facebook posts. As revealed, readability features of Flesch Kincaid reading ease and Gunning Fox indices, text length and number of hashtags are associated with both engagement and awareness metrics of likes, reaches and impressions.

Regarding RQ1, engagement metrics were associated with most of the examined readability features. The Gunning Fog index showed a significant association with engaged user and users who liked and engaged with the post, although no associations were detected for the Flesch Kincaid reading ease index. Regarding the text length attributes, both the number of words and characters indicated a positive association with engagement performance.

Regarding RQ2, both readability indexes revealed significant associations with the examined awareness metrics. Similarly, text length attributes and number of hashtags were positively associated with the awareness metrics.

Following we discuss the results for each one of the examined text characteristics

5.1. Text characteristics and user engagement

Regarding the readability indices, the Gunning fog index indicated significant impact on both engagement and awareness metrics, while the “Flesch Kincaid reading ease” index revealed statistically significant differences per reading ease category only in the metric of “Lifetime post impressions by people who have liked your page”. Referring to the categories: “Fairly easy to read” and “Easy to read” revealed the highest mean scores in “Lifetime post impressions by people who have liked your page”, and the category of "Difficult" revealed the lowest ones, even lower than category “Very difficult”. This outcome might provide some basic insights on the text data manipulation from the Facebook algorithm to ‘decide’ on the awareness performance of the posts.

As regards the Gunning fog index, differentiations were revealed in “Lifetime post engaged users”, “Lifetime post impressions by people who have liked your page”, “lifetime post reaches by people who like your page”, “Lifetime people who have liked your page and engaged with your post”.

The readability findings align with previous research (e.g., Davis et al., 2019) indicating the positive impact of readability on user engagement and brand awareness in social media posts. This study extends previous findings by revealing the value of readability in Facebook organic image posts. The current findings support that readability scores on text in image posts should receive higher consideration since they might associate with both user engagement and awareness metrics in Facebook organic image posts.

Regarding the text length, the results indicated that the posts containing more than 31 words, or more than 321 characters achieved the highest scores, while the lowest performance was achieved by the posts ranging up to 15 words or 160 characters. These finding are complementary to the study of Jackson (2020) who assessed the optimal number of characters and words in headlines, but not in the description text of image posts Davis et al. (2019). also supported the positive link between the text length and user engagement in tweets.

As regards the hashtag correlations with post performance metrics, the findings revealed that the number of hashtags correlates positively with all the examined Facebook performance metrics, suggesting that posts containing many hashtags tend to achieve higher performance scores. This result is confirming that the use of hashtags engage users throughout all social media networks (Cheng et al., 2020; Salomon, 2013), despite the fact that hashtags used to decrease the performance of Facebook posts when they were initially introduced in the Facebook platform. It seems that today, the wide use of other hashtag-rich social networks like Instagram and Twitter is associated with the way users perceive hashtag-rich content in Facebook.

5.2. Research limitations and future work

It is essential to pinpoint that this work brings certain limitations which hopelfully will be surpassed in the near future. To beign with, this work focused on certain industry sector and more specifically in women fashion. Future research can include a wider dataset regarding the fashion idustry and generalize the findings to a wider population of social media users.

Second, this research focused on organic (not paid) posts without including any advertising campaigns, which might bring different engagment outcomes and ftuture research shall be conducted to compare the findings between organic and paid Facebook posts in the Fashion industry.

Moreover, this research was based on a six-month rage dataset, from April to October. Future research should be conducted considering larger datasets, including the whole year range or festive seasons like Christmas.

Also, the exported dataset posed some restrictions since some basic engagement metrics (e.g. number of shares, numbe rof saves, and number of comments) were not in sufficent amount and were not included in the analysis. Further research should be conducted in the near future giving the opportunity to investigate more metrics of engagement.

Complementary the above, this study could be extended to consider other and/or external factors affecting consumers’ brand engagement including for instance spam emails sent from marketeers and companies (Batra, Jain, Tikkiwal & Chakraborty, 2021).

Further research should be conducted to investigate the directions and the deegree of influence as well as the direct or indirect effects of the measured readability constructs on user engagement and awarness metrics.

Finally, another interesting research direction is to apply text mining techniques to analyze sentiment or other psychological/behavioral elements in the post’s description text and explore its impact on user engagement. As a fact, text mining techniques are largely applied on social media datasets (Kumar, Kar & Ilavarasan, 2021). Topic modeling can also be useful to associate topics with user engagement levels in fashion related social media posts (Ahn et al., 2021).

5.3. Practical implications

This study focused on revealing the text readability factors that are associated with a set of performance metrics of engagement and awareness in Facebook organic image posts. The main contrition of the study is to provide deeper research insights on the factors associating with user engagement and awareness in social media branded messages. The results can be useful to marketers, business owners and decision makers assisting them to efficiently promote their businesses through organic posts and interpret the users' interactions. For instance, the study provides some key points and guidelines to managers who attempt to promote women fashion image content of Facebook using organic posts adding hashtags, keywords, long descriptions, and easy to read texts.

By revealing the essential role of the text in image posts, this study also contributes to previous literature and the research gap on consumer brand engagement in social media.

6. Conclusion

The study responds to the need to better conceive the factors that associate with consumer brand engagement in social media branded messages. Based on previous literature, this study focused on the examination of user behavior on branded image posts extracted from a Fashion retail Facebook business page. To fill the research gap in the research on the role of post characteristics on consumer engagement, this study examined a set of readability attributes including i) two popular readability indices (Flesch Kincaid reading ease and Gunning Fox), ii) the length of the text expressed in word length and characters length, and iii) the number of hashtags. The Facebook performance metrics considered a set of engagement and awareness metrics, including metrics of likes, reaches and impressions.

The statistical results indicated several significant associations between text readability features and performance metrics, highlighting the essential role of text on consumer engagement and awareness. Overall, the results revealed that, the posts' text which is easy to read, long (more than 31 words, or more than 321 characters), and contains many hashtags tends to achieve higher performance of engagement and awareness, expressed in higher values of "Lifetime post total reaches", "Lifetime Post Total Impressions" and "Lifetime post engaged users" variables. The most engaging Facebook posts tend to have the highest values in "Characters' count", "Word number", "Hashtags number" "Lifetime post total reaches", "Lifetime post total impressions" while they do not associate with readability metrics like the reading ease. Facebook posts with the highest "Characters count" tend to retain some of the highest values of the "Lifetime post total reaches", "Lifetime post total impressions", and "Lifetime post engaged users".

Overall, this study contributes to the global literature regarding consumer behavior and social media marketing, expanding in that way the limits of previous studies and presenting new findings that will allow stakeholders deeper understand consumer behavior and perform highly engaging social media marketing campaigns.

References

Ahn, J., Son, H., & Chung, A. D. (2021). Understanding public engagement on twitter using topic modeling: The 2019 Ridgecrest earthquake case. *International Journal of Information Management Data Insights, 1*(2), Article 100033. 10.1016/j.jjimei.2021.100033.

Antoniadis, I., Xanthakou, S., & Assimakopoulos, C. (2019). Consumers' perceptions and usage of social media in fashion marketing.

Batra, J., Jain, R., Tikkiwal, V. A., & Chakraborty, A. (2021). A comprehensive study of spam detection in e-mails using bio-inspired optimization techniques. *International Journal of Information Management Data Insights, 1*(1), Article 100006. 10.1016/j.jjimei.2020.100006.

Bianchi, C., & Andrews, L. (2018). Consumer engagement with retail firms through social media: An empirical study in Chile. *International Journal of Retail and Distribution Management, 46*(4), 364–385. 10.1108/IJRDM-02-2017-0035.

Cheng, W. W. H., Lam, E. T. H., & Chiu, D. K. W. (2020). Social media as a platform in academic library marketing: A comparative study. *Journal of Academic Librarianship, 46*(5), Article 102188. 10.1016/j.acalib.2020.102188.

Colladon, A. F., Gloor, P., & Iezzi, D. F. (2020). Editorial introduction: The power of words and networks. *International Journal of Information Management, 51*, Article 102031.

Consuegra, D. M., Faraoni, M., Díaz, E., & Ranfagni, S. (2018). Exploring relationships among brand credibility, purchase intention and social media for fashion brands: A conditional mediation model. *Journal of Global Fashion Marketing, 9*(3), 237–251. 10.1080/20932685.2018.1461020.

Cortés, J. D., Rivera, L., & Carbonelld, K. B. (2022). Mission statements in universities: Readability and performance. *European Research on Management and Business Economics, 28*(2), Article 100183. 10.1016/j.iedeen.2021.100183.

Cvijikj, I. P., Spiegler, E. D., & Michahelles, F. (2011). The effect of post type, category and posting day on user interaction level on Facebook. In *IEEE Third International Conference on Privacy, Security, Risk and Trust and IEEE Third International Conference on Social Computing* (pp. 810–813). 2011. 10.1109/PASSAT/SocialCom.2011.21.

Dale, E., & Chall, J. S. (1949). In *The Concept of Readability: 26* (p. 23). Elementary English.

Davenport, J.R.A., & DeLine, R. (2014). *The readability of tweets and their geographic correlation with education*. Retrieved from http://arxiv.org/abs/1401.6058

Davidaviciene, V., Davidavičius, S., & Tamosiuniene, R. (2019). B2C marketing communication in social media: Fashion industry specifics. In *International Conference on Creative Business for Smart and Sustainable Growth (CREBUS)* (pp. 1–4). 10.1109/CREBUS.2019.8840067.

Davis, S. W., Horváth, C., Gretry, A., & Belei, N. (2019). Say what? How the interplay of tweet readability and brand hedonism affects consumer engagement. *Journal of Business Research, 100*(February), 150–164. 10.1016/j.jbusres.2019.01.071.

Dhaoui, C., & Webster, C. M. (2021). Brand and consumer engagement behaviors on Facebook brand pages: Let's have a (positive) conversation. *International Journal of Research in Marketing, 38*(1), 155–175. 10.1016/j.ijresmar.2020.06.005.

Dong, W., Liao, S., & Zhang, Z. (2018). Leveraging financial social media data for corporate fraud detection. *Journal of Management Information Systems, 35*(2), 461–487.

Eleyan, D., Othman, A., & Eleyan, A. (2020). Enhancing software comments readability using Flesch Reading ease score. *Information, 11*(9), 430. 10.3390/info11090430.

Eslami, S. P., Ghasemaghaei, M., & Hassanein, K. (2021). Understanding consumer engagement in social media: The role of product lifecycle. *Decision Support Systems, xxxx*, Article 113707. 10.1016/j.dss.2021.113707.

Flesch, R. (1948). A new readability yardstick. *Journal of Applied Psychology, 32*(3), 221–233. 10.1037/h0057532.

Fournier, S., & Avery, J. (2011). The uninvited brand. *Business Horizons, 54*(3), 193–207. 10.1016/j.bushor.2011.01.001.

Gunning, R. (1969). The fog index after twenty years. *Journal of Business Communication, 6*(2), 3–13. 10.1177/002194366900600202.

Hsiao, S. H., Wang, Y. Y., Wang, T., & Kao, T. W. (2020). How social media shapes the fashion industry: The spillover effects between private labels and national brands. *Industrial Marketing Management, 86*(February 2019), 40–51. 10.1016/j.indmarman.2019.02.022.

Ismail, A., Kuppusamy, K. S., Kumar, A., & Ojha, P. K. (2019). Connect the dots: Accessibility, readability and site ranking – An investigation with reference to top ranked websites of Government of India. *Journal of King Saud University - Computer and Information Sciences, 31*(4), 528–540. 10.1016/j.jksuci.2017.05.007.

Jackson, D. (2020). Know your limit: The ideal length of every social media post. https://sproutsocial.com/insights/social-media-character-counter/ Accessed May 8, 2021

Kachba, Y. R., Plath, A. M. S., Ferreira, M. G. G., & Forcellini, F. A. (2012). Alignment of quality management in SMEs with the market performance. *Journal of Technology Management and Innovation, 7*(4), 103–111. 10.4067/s0718-27242012000400008.

Keyhole. (2020). Reaches vs impression: What's the difference and why is it important?. https://keyhole.co/blog/reaches-vs-impressions/ Accessed December 15, 2020

Khan, F., Hafeez, Z., Ijaz, A., & Shaheen, R. (2019). *Predicting Factors to Get Maximum Social Media Engagement of Customers for Brand Building*. Riyadh,Saudia Arabia: Al Yamamah University Engineering Forum (YUENG).

Klare, G. R. (1963). *The Measurement of Readability*. Ames, Iowa: Iowa State University Press.

Kumar, S., Kar, A. K., & Ilavarasan, P. V. (2021). Applications of text mining in services management: A systematic literature review. *International Journal of Information Management Data Insights, 1*(1), Article 100008. 10.1016/j.jjimei.2021.100008.

Kushwaha, A. K., & Kar, A. K. (2020a). Language Model-driven chatbot for business to address marketing and selection of products. In S. K. Sharma, Y. K. Dwivedi, B. Metri, & N. P. Rana (Eds.), *Re-imagining Diffusion and Adoption of Information Technology and Systems: A Continuing Conversation. TDIT 2020. IFIP Advances in Information and Communication Technology*: 617. Springer. 10.1007/978-3-030-64849-7_3.

Kushwaha, A. K., Kar, A. K., & Ilavarasan, P. V. (2020c). Predicting retweet class using deep learning. *Trends in Deep Learning Methodologies: Algorithms, Applications, and Systems*, 89.

Kushwaha, A. K., Mandal, S., Pharswan, R., Kar, A. K., & Ilavarasan, P. V. (2020b). Studying online political behaviours as rituals: A study of social media behaviour regarding the CAA. In S. K. Sharma, Y. K. Dwivedi, B. Metri, & N. P. Rana (Eds.), *Re-imagining Diffusion and Adoption of Information Technology and Systems: A Continuing Conversation. TDIT 2020. IFIP Advances in Information and Communication Technology*: 618. Springer. 10.1007/978-3-030-64861-9_28.

Lee, A. Y., & Aaker, J. L. (2004). Bringing the frame into focus: The influence of regulatory fit on processing fluency and persuasion. *Journal of Personality and Social Psychology, 86*(2), 205–218. 10.1037/0022-3514.86.2.205.

Lee, A. Y., & Labroo, A. A. (2004). The effect of conceptual and perceptual fluency on brand evaluation. *Journal of Marketing Research, 41*(2), 151–165. 10.1509/jmkr.41.2.151.28665.

Lee, K. (2020). The ideal length of everything online, backed by research. https://buffer.com/library/the-ideal-length-of-everything-online-according-to-science/ Accessed March 12, 2021

Leung, X. Y., Bai, B., & Erdem, M. (2017). Hotel social media marketing: A study on message strategy and its effectiveness. *Journal of Hospitality and Tourism Technology, 8*(2), 239–255. 10.1108/JHTT-02-2017-0012.

Molina, A., Gómez, M., Lyon, A., Aranda, E., & Loibl, W. (2020). What content to post? Evaluating the effectiveness of facebook communications in destinations. *Journal of Destination Marketing and Management, 18*(October). 10.1016/j.jdmm.2020.100498.

Pattnaik, S., & Trivedi, P. (2020). The impact of social media marketing on online fashion industry.

Posner, H. (2015). *Marketing Fashion: Strategy, Branding, and Promotion*. London: Laurence King Publishing.

Proserpio, D., & Zervas, G. (2017). Online reputation management: Estimating the impact of management responses on consumer reviews. *Marketing Science, 36*(5), 645–665. 10.1287/mksc.2017.1043.

Rennekamp, K. (2012). Processing fluency and investors' reactions to disclosure readability. *Journal of Accounting Research, 50*(5), 1319–1354. 10.1111/j.1475-679X.2012.00460.x.

Salomon, D. (2013). Moving on from Facebook: Using Instagram to connect with undergraduates and engage in teaching and learning. *College & Research Libraries News, 74*(8), 408–412 HTtps://doi.org/. 10.5860/crln.74.8.8991.

Santos, Z. R., Cheung, C., Coelho, P. S., & Rita, P. (2022). Consumer engagement in social media brand communities: A literature review. *International Journal of Information Management, 63*(November 2021). 10.1016/j.ijinfomgt.2021.102457.

Schulze, C., Schöler, L., & Skiera, B. (2014). Not all fun and games: Viral marketing for utilitarian products. *Journal of Marketing, 78*(1), 1–19. 10.1509/jm.11.0528.

Shapiro, S., & Wilk, M. (1965). An analysis of variance test for normality (Complete Samples). *Biometrika, 52*(3/4), 591–611. 10.2307/2333709.

Sreejesh, S., Paul, J., Strong, C., & Pius, J. (2020). Consumer response towards social media advertising: Effect of media interactivity, its conditions and the underlying mechanism. *International Journal of Information Management, 54*(June), Article 102155. 10.1016/j.ijinfomgt.2020.102155.

Sweeney, J., & Swait, J. (2008). The effects of brand credibility on customer loyalty. *Journal of Retailing and Consumer Services, 15*, 179–193. 10.1016/j.jretconser.2007.04.001.

Temnikova, I., Vieweg, S., & Castillo, C. (2015). The case for readability of crisis communications in social media. In *WWW 2015 Companion - Proceedings of the 24th International Conference on World Wide Web* (pp. 1245–1250). 10.1145/2740908.2741718.

The Design Space. (2020). How to get seen in the Facebook news feed. https://thedesignspace.co/how-to-get-seen-in-the-facebook-news-feed/ Accessed December 11, 2020

Wang, L. W., Miller, M. J., Schmitt, M. R., & Wen, F. K. (2013). Assessing readability formula differences with written health information materials: Application, results, and recommendations. *Research in Social and Administrative Pharmacy, 9*(5), 503–516. 10.1016/j.sapharm.2012.05.009.

WebFX. (2020). Readability test tool. Quick and easy way to test the readability of your work. https://www.webfx.com/tools/read-able/ Accessed October 1, 2020

Wu, P., Li, X., Shen, S., & He, D. (2020). Social media opinion summarization using emotion cognition and convolutional neural networks. *International Journal of Information Management, 51*(December 2018), Article 101978. 10.1016/j.ijinfomgt.2019.07.004.

Zhang, T., Lu, C., Torres, E., & Chen, P. J. (2018). Engaging customers in value co-creation or co-destruction online. *Journal of Services Marketing, 32*(1), 57–69. 10.1108/JSM-01-2017-0027.